

1. Preprocessing & Windowing

DASIG: 60 Subjects $\times$ 3 Trials
5 Wireless MIMUs (65 Channels) @ 200 Hz



Sliding Window: 0.5 s (100 steps, 50% overlap)
Per-Trial Z-Score Standardization



Window-Level Labeling: $y = \max(t_labels)$
69,094 Windows (87.6% SAFE / 12.4% DANGER)



2. Dynamic Kinematic Feature Expansion

Kinematic Channels ($3 \times 65 = 195$)

Position (Raw): $X \in \mathbb{R}^{65 \times 100}$



Velocity (1st Deriv): $V = dX/dt \in \mathbb{R}^{65 \times 100}$



Acceleration (2nd Deriv): $A = d^2X/dt^2 \in \mathbb{R}^{65 \times 100}$



Expanded Kinematic Tensor: [X; V; A]
Dimensions: (B, 195, 100)